

Weighted-Span Obstructions for Sparse Elastic Rectangle Frames on Velocity Grids

Runyuan Wang

Draft revised 21 August 2026

Abstract

Let $V_m = \{-m, \dots, m\}^3 \subset \mathbb{Z}^3$. An elastic rectangle gives a signed four-point row that annihilates the five collision modes $1, x, y, z$, and $|v|^2$. For a family of such rows, let C_m be the signed matrix, let D_m be the diagonal matrix of vertex incidence degrees, and use the canonical normalization $A_m = \frac{1}{2}C_mD_m^{-1/2}$. We give a self-contained weighted-span obstruction. If R has orthogonal side vectors p_R, q_R , then the least singular value of A_m on the quotient by the five weighted collision modes obeys

$$\gamma_m^2 \leq G_m^{-1} \sum_{R \in \mathcal{R}_m} (p_{R,x}q_{R,x} - p_{R,y}q_{R,y})^2 \leq G_m^{-1} \sum_{R \in \mathcal{R}_m} |p_R|^2 |q_R|^2,$$

where $G_m = \sum_{v \in V_m} (x^2 - y^2)^2$. Thus uniformly bounded degree and sublinear side lengths $L_m = o(m)$ cannot produce a uniform normalized gap. Conversely, any bounded-degree family with $\gamma_m \geq c > 0$ must put a positive fraction of rectangles at quantitatively macroscopic scale. This condition is necessary, not sufficient.

We separately report four finite full-pool residual-greedy instances for $m = 2, 3, 4, 5$. Exact odd-minor certificates show that each of their rational kernels is precisely the five-dimensional collision space, while numerical normalized gaps range from approximately 0.138 to 0.104. The exact kernel statements and the floating-point gaps are finite evidence only. Only the quantitatively macroscopic, diameter-scale regime remains open for the unrestricted nonlocal bounded-degree proposition.

1 Introduction and scope

This paper asks whether collision-type incidence structures can support sparse, well-conditioned constraint systems. The result proved here is a statement about finite matrices built from orthogonal velocity rectangles.

The paper has two deliberately separated components. First, we prove an explicit upper bound for the normalized quotient gap when both vertex degree and rectangle side length are bounded. This is an infinite-family theorem, but only under those two hypotheses. Second, we describe four finite matrices selected from complete rectangle pools. Their rational kernels are certified exactly; their gaps are computed numerically and independently recomputed from the raw row lists. Four finite cases do not establish a uniform asymptotic lower bound.

For clarity, the evidential status of assertions is as follows.

label	status	meaning in this paper
R	rigorous	derived symbolically for every object satisfying the stated hypotheses;
E	exact finite	certified over $\mathbb{Z}$ or $\mathbb{Q}$ for a named finite matrix;
N	numerical finite	floating-point computation with residual and reproducibility checks;
O	open	neither proved nor disproved by the available argument or data.

The weighted-span obstruction is R. The rational ranks and degree counts in Section 5 are E. The gaps in that section are N. The unrestricted nonlocal proposition in Section 8 is O.

2 Elastic rectangles and the normalized quotient

2.1 Rectangle rows

Fix an integer $m \geq 1$ and write

$$V_m = \{-m, -m+1, \dots, m\}^3, \quad N_m = |V_m| = (2m+1)^3.$$

An elastic rectangle in V_m is a quadruple that can be represented as

$$a, \quad a+p, \quad a+q, \quad a+p+q, \quad p, q \in \mathbb{Z}^3 \setminus \{0\}, \quad p \cdot q = 0, \quad (1)$$

with all four vertices in V_m . Its momentum identity is automatic. The energy residual equals

$$|a|^2 + |a+p+q|^2 - |a+p|^2 - |a+q|^2 = 2p \cdot q = 0.$$

For a scalar function $f : V_m \rightarrow \mathbb{R}$, define the row functional

$$\Delta_{a,p,q} f = f(a) + f(a+p+q) - f(a+p) - f(a+q). \quad (2)$$

Reordering the two diagonals changes at most the sign of this row and does not change its singular values.

Let $\mathcal{R}_m$ be a set of distinct elastic rectangles. The signed matrix $C_m \in \{-1, 0, 1\}^{|\mathcal{R}_m| \times N_m}$ has one row per rectangle, with the signs in (2). Each row has four nonzero entries. Its incidence degree at a vertex v is

$$d_v = |\{R \in \mathcal{R}_m : v \in R\}|, \quad D_m = \text{diag}(d_v : v \in V_m).$$

Throughout the normalized discussion we require $d_v > 0$ for every vertex. If a vertex is isolated, $D_m^{-1/2}$ is not defined in the intended normalization; moreover, the associated coordinate vector supplies an extra unconstrained direction in the unnormalized row matrix.

2.2 Collision modes and the canonical gap

Let

$$\mathcal{K}_m = \text{span}\{1, x, y, z, x^2 + y^2 + z^2\} \subset \mathbb{R}^{V_m}. \quad (3)$$

Direct substitution in (2) shows that $C_m k = 0$ for every $k \in \mathcal{K}_m$. The five functions are independent for $m \geq 1$. For instance, evaluating them at

$$(0, 0, 0), (1, 0, 0), (-1, 0, 0), (0, 1, 0), (0, 0, 1)$$

gives a 5×5 matrix whose determinant has absolute value 2.

We use the degree-normalized signed operator

$$A_m = \frac{1}{2} C_m D_m^{-1/2}. \quad (4)$$

The factor $1/2$ fixes the scale of a four-point row. In particular, the associated combinatorial convention is $4A_m^* A_m$. The known nullspace in normalized coordinates is $D_m^{1/2} \mathcal{K}_m$.

Definition 2.1 (Canonical quotient gap). The normalized gap is

$$\gamma_m = \min\{\|A_m X\|_2 : \|X\|_2 = 1, X \perp D_m^{1/2} \mathcal{K}_m\}. \quad (5)$$

Equivalently,

$$\gamma_m^2 = \frac{1}{4} \min_{f \notin \mathcal{K}_m} \frac{\sum_{R \in \mathcal{R}_m} |\Delta_R f|^2}{\min_{h \in \mathcal{K}_m} \sum_{v \in V_m} d_v |f(v) - h(v)|^2}. \quad (6)$$

The quotient in (5) removes only the five known modes. Thus an additional kernel vector makes $\gamma_m = 0$. Conversely, the exact identity $\ker_{\mathbb{Q}} C_m = \mathcal{K}_m$ for a finite rational matrix does not supply a scale-independent lower bound on γ_m .

3 The weighted-span obstruction

We now give the rigorous result of the paper. Vector lengths are Euclidean. For each rectangle R , choose one representation (1) and write its side vectors as p_R, q_R .

Proposition 3.1 (Weighted-span test). *Let $m \geq 1$ and suppose every vertex has positive incidence degree. Then the gap in Definition 2.1 satisfies*

$$\gamma_m^2 \leq G_m^{-1} \sum_{R \in \mathcal{R}_m} (p_{R,x} q_{R,x} - p_{R,y} q_{R,y})^2 \leq G_m^{-1} \sum_{R \in \mathcal{R}_m} |p_R|^2 |q_R|^2, \quad (7)$$

where

$$G_m = \sum_{(x,y,z) \in V_m} (x^2 - y^2)^2.$$

Writing $S_j = \sum_{t=-m}^m t^j$, one has

$$G_m = 2S_0(S_0 S_4 - S_2^2) \quad (8)$$

$$= \frac{2(2m+1)^3 m(m+1)(4m^2 + 4m - 3)}{45}. \quad (9)$$

Proof. Set

$$g(x, y, z) = x^2 - y^2.$$

Cubic symmetry makes g orthogonal, in the unweighted counting inner product on V_m , to all of $\mathcal{K}_m$. Indeed, exchanging x and y handles the constant and radial quadratic modes, while sign symmetry handles the three linear modes.

Use the degree-weighted inner product

$$\langle u, w \rangle_d = \sum_{v \in V_m} d_v u(v) w(v), \quad \|u\|_d^2 = \langle u, u \rangle_d.$$

Let $h \in \mathcal{K}_m$ be the degree-weighted orthogonal projection of g onto $\mathcal{K}_m$, and put $f = g - h$. Then $X = D_m^{1/2}f$ is orthogonal to $D_m^{1/2}\mathcal{K}_m$. Also, $C_m h = 0$, so $C_m f = C_m g$. The denominator has the lower bound

$$\begin{aligned} \|X\|_2^2 &= \|f\|_d^2 = \min_{k \in \mathcal{K}_m} \sum_v d_v |g(v) - k(v)|^2 \\ &\geq \min_{k \in \mathcal{K}_m} \sum_v |g(v) - k(v)|^2 = \|g\|_2^2 = G_m. \end{aligned} \quad (10)$$

Here the inequality uses $d_v \geq 1$, and the final equality uses the unweighted orthogonality of g to $\mathcal{K}_m$. In particular, X is nonzero.

For a rectangle represented by (a, p, q) , direct expansion gives

$$\Delta_{a,p,q}g = 2(p_x q_x - p_y q_y). \quad (11)$$

Using $A_m X = \frac{1}{2}C_m f = \frac{1}{2}C_m g$ and (11), we obtain

$$\|A_m X\|_2^2 = \frac{1}{4} \sum_{R \in \mathcal{R}_m} |\Delta_R g|^2 = \sum_{R \in \mathcal{R}_m} (p_{R,x} q_{R,x} - p_{R,y} q_{R,y})^2. \quad (12)$$

The Rayleigh quotient of this admissible X , together with (10), proves the first inequality in (7). The second follows from Cauchy–Schwarz:

$$|p_x q_x - p_y q_y| \leq \sqrt{p_x^2 + p_y^2} \sqrt{q_x^2 + q_y^2} \leq |p| |q|.$$

It remains to evaluate G_m . Expanding and summing first in x, y and then in z yields

$$G_m = S_0(2S_0 S_4 - 2S_2^2),$$

which is (8). The standard finite sums are

$$S_0 = 2m + 1, \quad S_2 = \frac{m(m+1)(2m+1)}{3}, \quad S_4 = \frac{m(m+1)(2m+1)(3m^2+3m-1)}{15}.$$

Substitution gives (9). □

Corollary 3.2 (Sublinear spans give no uniform gap). *Let $(\mathcal{R}_m)$ be a family for unbounded m . Suppose every vertex has positive degree, $d_v \leq D$ with D independent of m , and every rectangle has $|p_R|, |q_R| \leq L_m$. Then*

$$\gamma_m \leq \frac{L_m^2}{2} \sqrt{\frac{45D}{2m(m+1)(4m^2+4m-3)}} = O_D((L_m/m)^2). \quad (13)$$

In particular, $L_m = o(m)$ implies $\gamma_m \rightarrow 0$. Fixed-span families have $L_m = O(1)$ and recover the older $O_D(m^{-2})$ local obstruction. More generally, any sublinear-span family fails to be a uniform expander under (4)–(5). The only span regime not excluded by this test is diameter scale, $L_m = \Theta(m)$.

Proof. Every row has four distinct vertices, so $4|\mathcal{R}_m| = \sum_v d_v \leq D(2m+1)^3$. Combining this count with Proposition 3.1 and $|p_R|, |q_R| \leq L_m$ gives

$$\gamma_m^2 \leq \frac{D(2m+1)^3 L_m^4}{4G_m}.$$

Substituting (9) gives (13). □

Corollary 3.3 (Macroscopic-span necessary condition). *Fix constants $D < \infty$ and $c > 0$. Let*

$$\rho = \min\{1, c/\sqrt{135D}\}, \quad \eta = \min\{1, c^2/(540D)\}.$$

If a family on V_m has $1 \leq d_v \leq D$ and $\gamma_m \geq c$, then at least an η fraction of its rectangles simultaneously satisfy

$$|p_R| \geq \rho m, \quad |q_R| \geq \rho m.$$

Thus a scale-independent gap forces a positive density of quantitatively macroscopic, diameter-scale rectangles. This condition is necessary only; it is not sufficient for a gap.

Proof. From (9), $G_m \geq (4/45)(2m+1)^3 m^4$. Also $|\mathcal{R}_m| \leq D(2m+1)^3/4$. In the cube V_m , every side vector has length at most $2\sqrt{3}m$, so $|p_R|^2 |q_R|^2 \leq 144m^4$.

Call a rectangle bad if at least one of its two side lengths is $< \rho m$. For a bad rectangle, $|p_R|^2 |q_R|^2 \leq 12\rho^2 m^4$. Therefore the total bad contribution is at most $3D\rho^2(2m+1)^3 m^4 \leq (c^2/45)(2m+1)^3 m^4$. Since $\gamma_m \geq c$, Proposition 3.1 forces

$$\sum_R |p_R|^2 |q_R|^2 \geq c^2 G_m \geq (4c^2/45)(2m+1)^3 m^4.$$

The good rectangles therefore contribute at least $(c^2/15)(2m+1)^3 m^4$. Each contributes at most $144m^4$, so their number is at least $c^2(2m+1)^3/2160$. Dividing by $|\mathcal{R}_m| \leq D(2m+1)^3/4$ gives a fraction at least $c^2/(540D)$, and hence at least η . $\square$

Remark 3.4 (What remains open). Corollary 3.2 rules out fixed, local, block-glued, and all sublinear-span bounded-degree families under the stated normalization. Corollary 3.3 says any positive answer to the nonlocal problem must place a positive fraction of rows in the diameter-scale regime. The paper does not prove that such a macroscopic family exists, nor that every macroscopic family has a gap.

4 Canonical finite pools and exact kernel certificates

The finite experiment used all nonperiodic rectangles in V_m as its candidate pool. An unordered diagonal is a pair $\{u, w\} \subset V_m$. Diagonals were grouped by

$$(u+w, |u-w|^2).$$

Two distinct diagonals in one group have a common midpoint and equal length; their four endpoints form an elastic rectangle. Sorting the pairs and assigning $+, +$ to the earlier diagonal and $-, -$ to the later one produces a canonical row. Pairing every two distinct diagonals in every group gives the complete finite pool under this convention. Shared-vertex degeneracy cannot occur between two distinct pairs in one group: equal sums would force the other endpoints to coincide. Periodic boundaries were not used, because reducing coordinates modulo a period would not preserve the quadratic energy mode on the displayed representatives.

The designated finite instances were selected by a residual-greedy branch from these complete pools. The branch begins with a degree-capped balanced selection and then adds rows having large residual in the current numerical quotient computation. This is a search heuristic, not an infinite recursive construction and not an exhaustive optimization.

For each designated candidate, the exact certificate performs the following checks.

1. Every row is unique, belongs to the canonical pool, and satisfies the integer momentum, energy, and orthogonality identities.
2. All five integer mode columns in (3) have zero row residual, and their anchor determinant has absolute value 2. Therefore $\text{rank}_{\mathbb{Q}} C_m \leq N_m - 5$.
3. The certificate lists row and column indices of an $(N_m - 5) \times (N_m - 5)$ integer minor whose determinant is 1 modulo 2. The integer determinant is therefore nonzero, so $\text{rank}_{\mathbb{Q}} C_m \geq N_m - 5$.

It follows exactly, for that finite matrix, that

$$\text{rank}_{\mathbb{Q}} C_m = N_m - 5, \quad \ker_{\mathbb{Q}} C_m = \mathcal{K}_m.$$

Ranks over the two large primes 2147483647 and 2147483629 were also computed and equal $N_m - 5$. Those prime computations are independent screens; the odd maximal minor, not the prime screens, is the rational-rank proof.

5 Four finite full-pool instances

Table 1 records the four designated residual-greedy instances. The vertex and row counts, degree range, rational rank, and kernel dimension are exact finite data (E). The final column is a floating-point normalized gap (N), freshly recomputed from the raw JSON row list with the normalization (4). Digits shown are rounded and are not certified intervals.

Table 1: Finite residual-greedy instances. Columns through $\dim \ker_{\mathbb{Q}}$ are exact finite data; γ_m is numerical.

m	N_m	$ \mathcal{R}_m $	degree range	$\text{rank}_{\mathbb{Q}} C_m$	$\dim \ker_{\mathbb{Q}} C_m$	normalized γ_m
2	125	188	4–10	120	5	0.137567327104
3	343	515	3–10	338	5	0.108836005898
4	729	1094	3–10	724	5	0.107304440463
5	1331	1997	3–10	1326	5	0.104461520420

The independent numerical recomputation rebuilt C_m , D_m , A_m , and the weighted constraint basis directly from the four row arrays. A fresh constrained eigensolve gave, respectively,

$$0.13756732710431338, \quad 0.10883600589808418, \quad 0.10730444046255100, \quad 0.10446152042018168.$$

The fresh quotient-operator residuals were all below 6×10^{-10} , and no solver warning was emitted in those runs. These checks support the reported finite approximations. They do not turn the gaps into exact algebraic values.

Two non-implications are essential. First, exact five-dimensional kernels at four sizes do not imply exact kernels at later sizes. Second, the observed numbers do not imply that $\liminf_{m \rightarrow \infty} \gamma_m > 0$. The candidates were chosen from full, nonperiodic pools without a fixed side-length cutoff, so Corollary 3.2 does not itself rule them out; conversely, the four candidates do not define or certify an infinite nonlocal family.

6 Validation and reproducibility

The evidence package separates symbolic, exact finite, and numerical checks. The following layers were used.

1. **Formula and pool checks.** Integer arithmetic verifies every canonical row, hashes the ordered row arrays with domain-separated SHA-256, and rejects duplicate rows.
2. **Exact kernel checks.** The odd-minor procedure described in Section 4 was rerun independently for all four designated candidates. Fresh proof payloads agree with the packaged payloads after excluding run metadata.
3. **Raw-gap recomputation.** The normalized operators and constraint bases were reconstructed from the raw row arrays rather than from summary tables. Stored numerical vectors and fresh constrained eigenvectors were checked for norm, constraint residual, operator residual, and relative gap agreement.
4. **Theorem arithmetic checks.** Exact integer summation checks (8) and the five orthogonality identities for $m = 1, \dots, 10$. On four finite local block candidates, a separate script checks the identity (11), the side-length bound, the projected Rayleigh quotient, and the inequality chain in Proposition 3.1.
5. **Manuscript-local checks.** The draft package contains a source and claim checker, build logs, and a source matrix mapping each strong claim to an exact local artifact or to one of the two verified citations.

The finite checks in items 3 and 4 are validation aids. The general theorem depends on the proof in Section 3, not on extrapolation from those checks.

7 Literature-boundary checks

This revision made two targeted public literature checks. First, in discrete velocity models for the Boltzmann equation, the standard physical invariants are mass, momentum, and energy; extra invariants are called spurious or nonphysical, and DVMs without them are called normal [2, 1]. This is close in vocabulary to the finite statement $\ker C = \text{span}\{1, v_x, v_y, v_z, |v|^2\}$, but the checked sources concern kinetic DVM construction and mixtures, not this paper’s finite cubic-lattice rectangle-frame quotient gap.

Second, the locality obstruction is consistent with standard finite-state Poincaré and spectral-gap intuition: bounded-range walks on long one-dimensional or lattice-like sets have gaps that decay with the diameter; for example, the simple random walk on the n -cycle has spectral gap $1 - \cos(2\pi/n)$ [3]. This paper’s proof is nevertheless a separate test-function argument for the specific four-point collision operator, five-mode quotient, $1/2$ normalization, and constants above.

Claim	closest prior result checked	same/different	safe wording
Five physical collision modes	normal DVMs have no spurious collision invariants beyond mass, momentum and energy	same invariant vocabulary; different finite matrix and no gap theorem	exact finite kernels here are normal-DVM-like evidence, not a novelty or priority claim
Locality obstructs uniform gap	cycle/lattice spectral-gap decay under local moves	same Poincaré intuition; different operator and quotient	the paper proves a source-bound weighted-span obstruction for this model
Finite $m = 2, \dots, 5$ certificates	no direct finite cubic elastic-rectangle coverage found in the targeted check	related only by invariant language	certificates retain independent finite meaning; no uniform-gap evidence

8 The remaining nonlocal problem

A natural grid form of the unresolved question is the following.

Open nonlocal rectangle-frame proposition. Do there exist absolute constants $D < \infty$ and $c > 0$, and elastic-rectangle families on velocity grids V_m for unbounded m , such that every vertex has degree between 1 and D , $\ker_{\mathbb{Q}} C_m = \mathcal{K}_m$, and the normalized gap satisfies $\gamma_m \geq c$, when rectangle side lengths are allowed to grow with m ?

Corollary 3.3 identifies a necessary escape from locality: any positive answer must use a positive fraction of rectangles whose two side lengths are both bounded below by a fixed multiple of m (or must change another stated hypothesis). This is only a necessary condition. The four values in Table 1 neither prove nor disprove the open proposition. The broader existence question on other growing rational or integer velocity sets is likewise not settled by this draft.

9 Limitations and next technical step

The main limitations are structural rather than cosmetic.

- The theorem is one-sided and conditional on positive bounded degree. It rules out sublinear span but supplies no lower bound and no sharpness result.
- The exact certificates concern four finite matrices only. They do not provide a recursive construction.
- The spectral values are floating-point estimates, not interval or symbolic eigenvalue certificates.
- The computational search and attacks are finite. They are not claimed to cover all selection mechanisms, all nonlocal patterns, or all velocity sets.
- The two targeted literature checks in Section 7 are not a systematic priority review of rectangle-frame Poincaré inequalities. This paper therefore makes no novelty, priority, “current best,” or no-precedent claim.

The exact next mathematical step is to prove or refute a scale-free Poincaré inequality for an explicit nonlocal bounded-degree rectangle family, with the five collision modes as the full rational

kernel. A positive route needs a recursive construction and a uniform lower bound; a negative route needs a test function or decomposition that remains effective in the quantitatively macroscopic side-length regime. Extending the finite table alone is not a substitute for either argument. Before formal submission, a systematic literature review and independent mathematical review are also required.

10 Conclusion

The rigorous conclusion is narrow and explicit: on V_m , the weighted-span test rules out uniformly bounded-degree families whose rectangle side lengths are $o(m)$. A scale-independent gap would require a positive fraction of diameter-scale rectangles. Exact finite certificates at $m = 2, 3, 4, 5$ establish five-dimensional rational kernels for four selected matrices, and numerical recomputation supports their displayed finite gaps. Those observations do not settle the unrestricted nonlocal proposition, which remains open.

AI authorship, assistance, and responsibility disclosure

Every sentence of manuscript prose and the complete manuscript structure were generated by LingTai AI under Runyuan Wang’s direction and authorization. LingTai AI also performed source organization, mathematical and computational analysis assistance, and code-assisted validation. The original conjectural agenda and experimental direction came from Runyuan Wang. AI production is not evidence for any claim: the stated proofs, source-bound checks, finite certificates, and disclosed numerical computations must support the paper on their own. Independent human review remains mandatory before external use or submission. Runyuan Wang retains final scientific judgment and responsibility for any public or submitted version. This work is shared purely for personal enjoyment and playful exploration with AI. If anything is incorrect, please excuse it; criticism and corrections are warmly welcomed.

References

- [1] Niclas Bernhoff and Mirela Vinerean. Discrete velocity models for mixtures without nonphysical collision invariants. *Journal of Statistical Physics*, 165:201–219, 2016.
- [2] Alexander V. Bobylev and Carlo Cercignani. Discrete velocity models without nonphysical invariants. *Journal of Statistical Physics*, 97:677–686, 1999.
- [3] Sharad Goel, Ravi Montenegro, and Prasad Tetali. Mixing time bounds via the spectral profile. *Electronic Journal of Probability*, 11:1–26, 2006.